Multisim 14 · Rev C 完整搭建副本

基准：用户上传的GBUCK_STEP02_5R_U3_IIB_SMOOTH_100ms_4CH_FROM0.ms14；对照已修订的Rev C原理图搭建。原文件原字节归档。本次原生工程有113项主板器件、6项仿真夹具、1个接地参考。

主工程：GBUCK_REVC_FULL_25V_5R_100ms.ms14。25V输入、5Ω负载、30kHz控制，0 – 100ms、最大步长50ns；保留上传工程的求解容差，默认只记录VOUT、I(RS_OUT)、FAULT_N、IOUT_OC_N四路。

短检查：GBUCK_REVC_FULL_SMOKE_10ms.ms14。先用它检查本机打开、模型载入与启动；10ms仍在软启动阶段，不能按5V稳态验收。

预充检查：GBUCK_REVC_PRECHARGE_ONLY_110ms.ms14。输入从0升至25V，22Ω预充，100ms时模拟闭合旁路；MCU_EN始终0，驱动保持关闭。主工程则假设输入已预充、旁路已闭合，避免把100ms预充塞入同一个100ms控制试验。

搭建内容：IR2184驱动与4.7Ω栅极电阻、QL并联SS56、1Ω模拟供电、五路ADC滤波/钳位、±2.2A电感保护、1.10A独立输出告警、上电复位、锁存/重新ARM及控制接口均有可见编号引脚与真实原生网络。

图形说明：器件按功能区重排。保留R/C/L、MOS、二极管、电源及保险丝的原生标准符号；新增IC和接口采用编号方框。连线用同名网络短引线相连。新增IC属于用户自定义功能元件，不是Multisim自带的厂家认证器件。

验证边界：已通过双DCL解码与原生连接反查。新增IR2184、LM393、74HC和MCU为功能等效模型；三路采样保留上传工程内的模型，包括INA240的既有IIB平滑修改。未执行Windows Multisim客户端打开/运行、ERC/DRC或STM32固件测试。

全部模型内嵌在.ms14中；不需要另行安装元器件库或外置.lib文件。日期：2026-10-26（北京时间）

使用顺序、实物影响和验证状态

1. 在Multisim 14中打开10ms短检查副本，检查页面是否出现全部功能区、模型缺失或连接错误；运行Transient Analysis。先看FAULT_N、DRV_EN、VOUT和采样信号，再打开100ms主工程。原生客户端是否通过仍需本机确认。

2. 主工程启动：约3ms发ARM脉冲，3.2ms开放MCU_EN，4 – 24ms软启动；比较器过流/过压拉低FAULT_N并锁存。IOUT_OC_N只作为输出告警，功能MCU在告警时抑制高边脉冲，不直接清除硬件锁存。

3. Q1/Q2重命名为QH/QL。U1物理脚号改为1 IN、2 /SD、3 COM、4 LO、5 VCC、6 VS、7 HO、8 VB。器件、排线和PCB均受影响；J_CTL.6改接IOUT_OC_N/PB11，不能沿用旧排线。

4. 实物预充时，外置旁路开关初始断开，至少100ms并确认VIN接近输入后再闭合；随后才允许ARM/启动。停机先断电放电，再打开旁路。J_BYP不是自动控制器，不能在带载运行中打开。

5. V_AUX12、V_AUX5、V_CORE3V3是明确标注的理想辅助电源夹具。它们没有从VIN汲取真实模块输入功率；不得根据此工程计算或宣称整机效率达85%。SIM_MCU是连续控制等效，不含12位量化、真实采样/中断延迟或可烧录固件。

6. 输出电容放电仍可能瞬时超过1A；本工程未加入输出串联硬限流。R_SN/C_SN为默认不装的调试支路，其符号/网络保留，电气模板不盖章到仿真电路。F1保持2A电气参数；跨引擎检查仅用1 mΩ导通态，不验证热熔断。

静态引脚反查：PASS_NATIVE_STATIC_READBACK。

10ms全电路跨引擎：完成。

100ms全电路跨引擎：完成。

110ms预充跨引擎：完成。

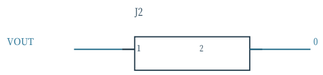
100ms最后5ms平均：VOUT=4.986405V，IOUT=1.000068A；FAULT_N正常。

跨引擎结果来自ngspice 44.2，不能等同Multisim本机运行或真实硬件验证；本机异常请保留原生错误日志。

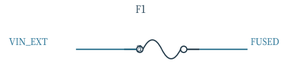
01 主功率级 / 15 – 25V → 10V, 1A

交付.ms14原生记录预览 | 同名网络相连 | 标准符号与功能IC方框的物理脚号、网络和电气模型均可编辑

2P POWER

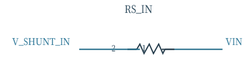


2P POWER



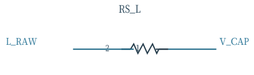
2A / 5x20

SS56

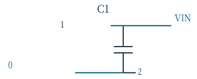


50mohm / 1% / >=1W

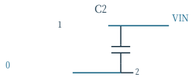
50mohm / 1% / >=1W



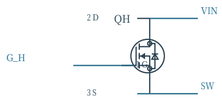
25mohm / 1% / >=1W



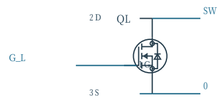
470uF / 50V



1uF / 50V



IRF540N



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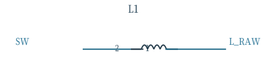
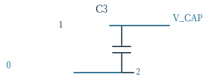
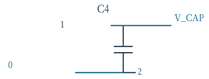


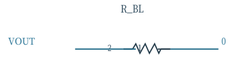
220uH / Isat>=3A



470uF / 35V



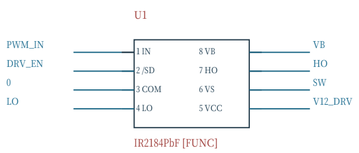
10uF / 35V

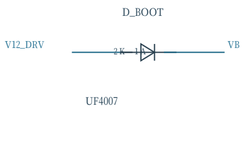


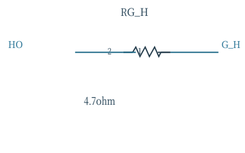
2.2kohm / 0.25W

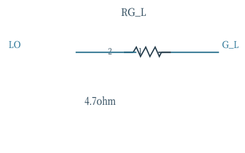
02 IR2184 单路PWM驱动与开关辅助电源

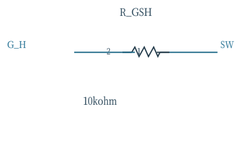
交付.ms14原生记录预览 | 同名网络相连 | 标准符号与功能IC方框的物理脚号、网络和电气模型均可编辑

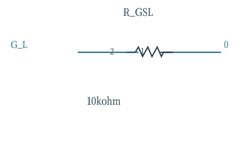


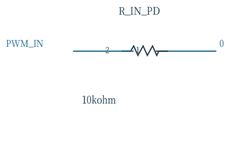


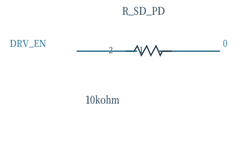


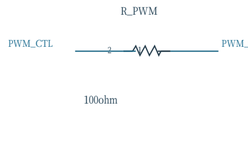


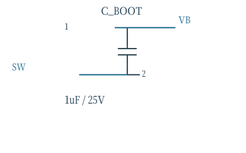


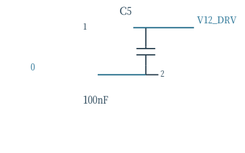


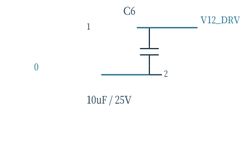


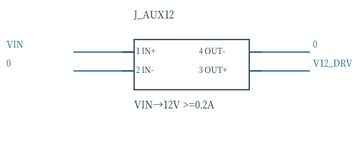


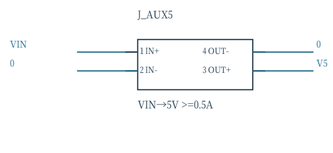






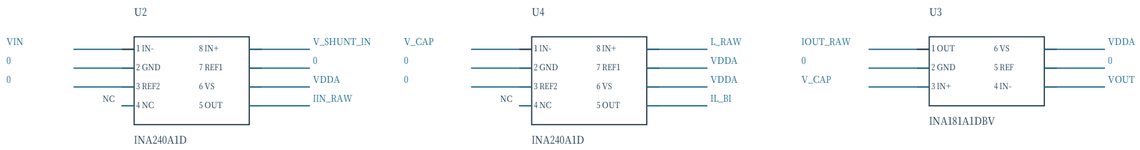


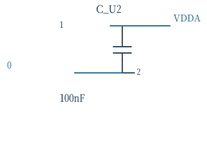


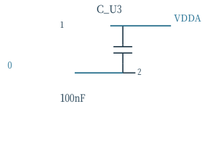


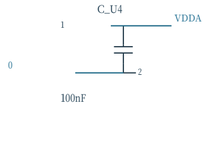
03 电流采样与模拟电源

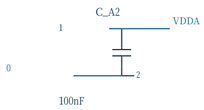
交付.ms14原生记录预览 | 同名网络相连 | 标准符号与功能IC方框的物理脚号、网络和电气模型均可编辑

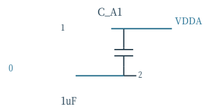


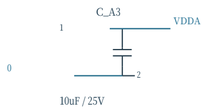






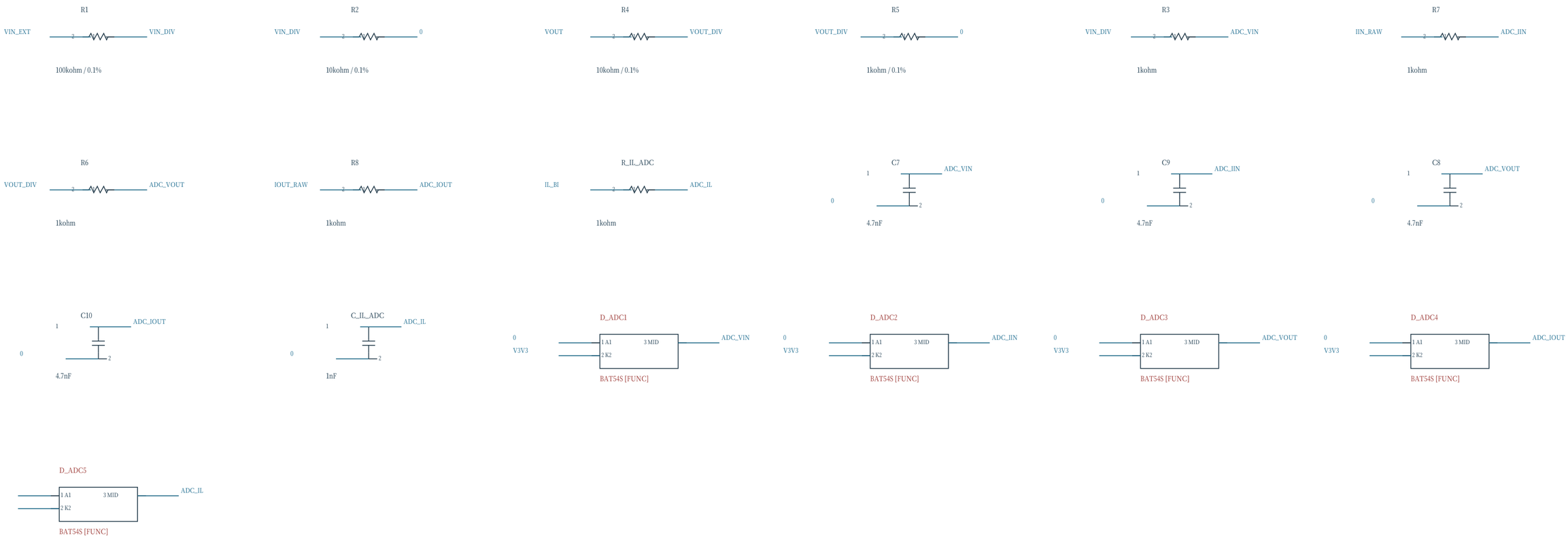






04 电压分压、五路ADC滤波与钳位

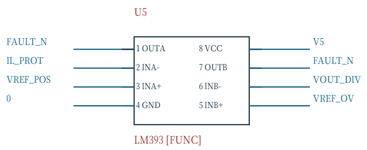
交付.ms14原生记录预览 | 同名网络相连 | 标准符号与功能IC方框的物理脚号、网络和电气模型均可编辑

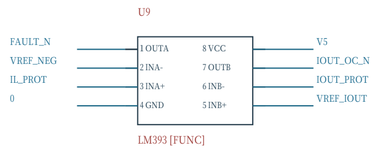


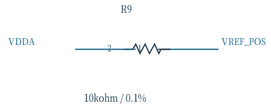
Rev C / 原生搭建预览 / 未执行Multisim客户端打开及运行 / GBUCK_REVC_FULL_25V_5R_100ms.ms14

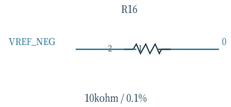
05 ±2.2A电感保护、1.10A输出告警及过压锁存

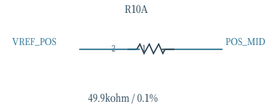
交付.ms14原生记录预览 | 同名网络相连 | 标准符号与功能IC方框的物理脚号、网络和电气模型均可编辑

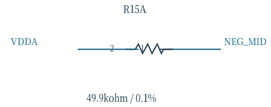


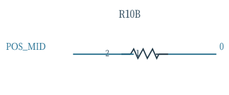


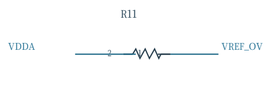


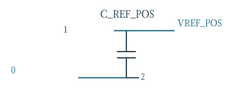


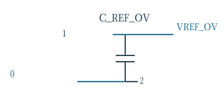


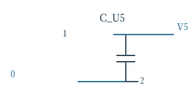


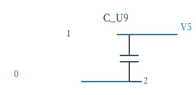






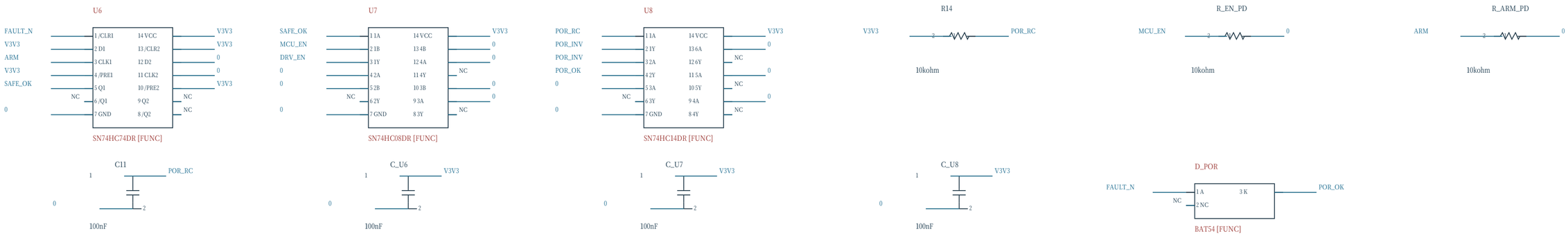






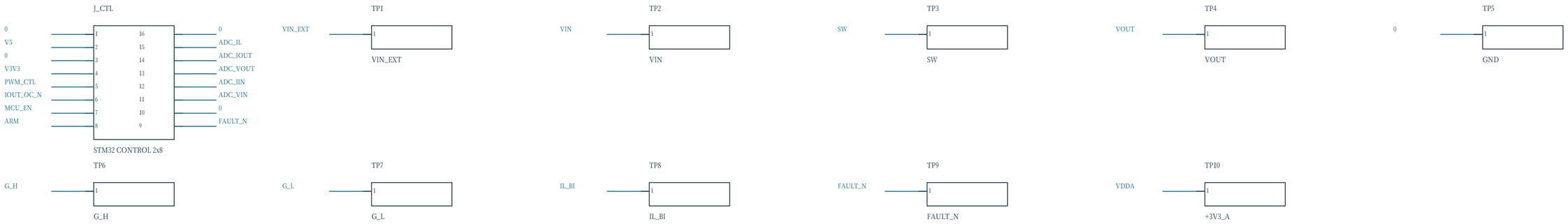
06 上电复位、故障锁存与使能

交付.ms14原生记录预览 | 同名网络相连 | 标准符号与功能IC方框的物理脚号、网络和电气模型均可编辑



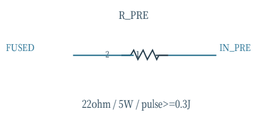
07 STM32控制板接口与测试点

交付.ms14原生记录预览 | 同名网络相连 | 标准符号与功能IC方框的物理脚号、网络和电气模型均可编辑

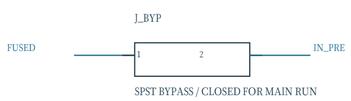


08 预充、续流、保护滤波与新增测试点

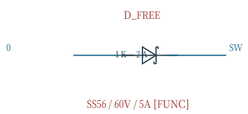
交付.ms14原生记录预览 | 同名网络相连 | 标准符号与功能IC方框的物理脚号、网络和电气模型均可编辑



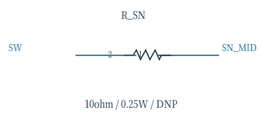
22ohm / 5W / pulse>=0.3J



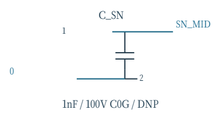
SPST BYPASS / CLOSED FOR MAIN RUN



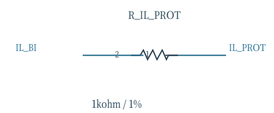
SS56 / 60V / 5A [FUNC]



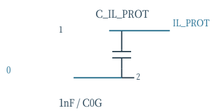
10ohm / 0.25W / DNP



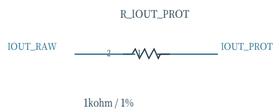
1nF / 100V C0G / DNP



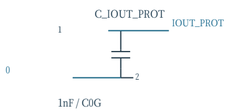
1kohm / 1%



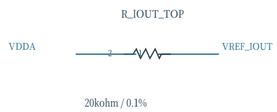
1nF / C0G



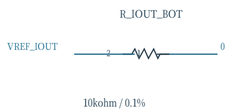
1kohm / 1%



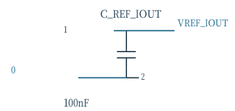
1nF / C0G



20kohm / 0.1%



10kohm / 0.1%



100nF

10kohm

1uF / 25V X7R

IOUT_RAW

IL_PROT

VREF_IOUT

09 仿真夹具与接地参考

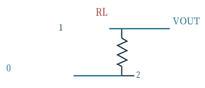
交付.ms14原生记录预览 | 同名网络相连 | 标准符号与功能IC方框的物理脚号、网络和电气模型均可编辑

25V ideal

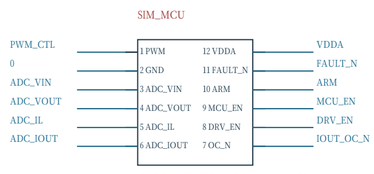
12V ideal

5V ideal

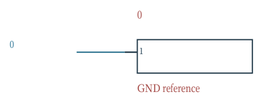
3.3V ideal



5ohm



CVCC 30kHz [EQ / NOT FIRMWARE]



GND reference